

BANATSKI KARLOVAC, Serbia

WMO: 131800

Lat: 45.050N

Lon: 21.033E

Elev: 91

StdP: 100.24

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.1	-13.5	1.2	-9.4	-10.4	1.6	-6.9	16.3	0.9	14.6	2.6	2.6	340	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.7	21.3	32.2	21.1	30.6	20.4	23.3	29.6	22.4	29.0	21.5	28.1	2.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.5	16.4	26.1	20.5	15.3	24.7	19.6	14.5	23.6	69.8	29.4	66.0	29.1	63.0	28.1	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.5	8.8	DB	-15.0	35.8	4.5	1.8	-18.2	37.1	-20.8	38.2	-23.3	39.2	-26.6	40.5	(4)
(5)			WB	-15.2	24.9	4.3	1.5	-18.4	26.0	-20.9	26.8	-23.3	27.6	-26.4	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	0.9	2.8	7.3	12.9	17.5	20.7	22.4	22.6	17.9	12.4	7.6	2.0	(6)	
	DBStd	8.81	5.05	5.41	4.77	4.06	3.74	3.47	2.97	3.12	3.79	4.55	4.76	4.80	(7)	
	HDD10.0	999	283	207	109	19	1	0	0	0	29	102	249		(8)	
	HDD18.3	2642	541	436	342	168	61	17	4	54	187	323	506		(9)	
	CDD10.0	1837	1	4	26	106	235	322	385	389	236	104	29	2	(10)	
	CDD18.3	438	0	0	0	4	36	88	131	136	39	4	0	0	(11)	
	CDH23.3	4494	0	0	1	56	343	900	1378	1397	376	44	0	0	(12)	
	CDH26.7	1690	0	0	0	10	73	319	568	603	113	3	0	0	(13)	
(14) Wind	WSAvg	3.6	4.0	4.1	4.3	4.3	3.6	2.9	2.6	2.5	3.0	3.5	4.3	3.9	(14)	
(15) Precipitation	PrecAvg	641	38	37	38	52	63	75	78	56	60	51	45	48	(15)	
	PrecMax	936	77	77	100	109	127	172	267	197	175	126	128	128	(16)	
	PrecMin	310	7	2	7	2	20	17	3	2	4	1	0	5	(17)	
	PrecStd	145	20	21	24	28	27	40	62	42	38	34	27	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	17.8	22.2	28.1	30.7	33.9	35.9	36.5	33.0	26.8	21.2	14.1	(19)	
		MCWB	9.6	10.4	12.2	16.3	18.6	21.6	21.8	21.6	19.7	17.2	13.9	10.6	(20)	
	2%	DB	10.9	14.6	19.6	24.7	28.7	32.1	33.5	34.1	29.7	24.6	18.6	12.0	(21)	
		MCWB	8.1	9.5	11.2	14.6	18.0	21.3	21.7	21.1	18.6	16.2	13.1	9.3	(22)	
	5%	DB	9.1	12.0	17.1	22.4	27.1	30.3	31.8	32.2	27.2	22.3	16.3	10.0	(23)	
		MCWB	7.2	8.5	10.3	13.5	17.4	20.6	21.4	20.9	17.9	15.1	12.0	7.9	(24)	
	10%	DB	7.4	9.9	14.7	20.3	25.2	28.5	30.0	30.2	25.1	19.9	14.3	8.2	(25)	
		MCWB	6.0	7.2	9.3	12.6	16.9	19.8	20.7	20.2	17.0	14.4	11.0	6.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.7	13.4	17.0	20.6	24.5	24.6	24.7	21.1	18.2	14.9	11.5	(27)	
		MCDB	12.4	15.8	19.7	24.1	26.3	29.8	30.4	30.3	28.8	23.4	19.5	13.2	(28)	
	2%	WB	8.6	10.0	12.0	15.6	19.5	22.8	23.3	23.1	19.8	17.0	13.7	9.3	(29)	
		MCDB	10.6	13.8	17.5	22.3	25.3	28.6	29.8	29.5	26.7	22.9	17.4	11.5	(30)	
	5%	WB	7.3	8.7	10.9	14.5	18.6	21.7	22.4	22.0	18.8	15.9	12.4	8.1	(31)	
		MCDB	8.9	11.7	16.0	20.4	24.3	27.9	29.1	29.0	24.8	20.6	15.5	9.7	(32)	
	10%	WB	6.1	7.5	9.9	13.6	17.8	20.7	21.6	21.1	17.9	15.0	11.3	6.8	(33)	
		MCDB	7.3	9.8	14.1	18.7	23.1	26.2	28.1	28.0	23.2	19.3	14.1	8.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	7.0	9.3	10.3	10.7	11.2	12.1	12.4	10.9	10.1	7.5	5.4	(35)	
		MCDBR	8.0	10.7	13.0	14.0	13.7	14.4	15.3	15.8	14.5	13.5	10.3	7.9	(36)	
	5% WB	MCWBR	5.6	6.3	6.6	6.0	4.9	5.4	5.6	5.3	5.4	6.0	5.2	5.4	(37)	
		MCWBR	7.5	9.4	11.1	11.7	11.8	12.8	13.3	13.2	12.2	11.9	9.5	7.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.371	0.404	0.460	0.447	0.441	0.447	0.459	0.434	0.396	0.361	0.343	(40)	
		taud	2.328	2.245	2.208	2.065	2.166	2.219	2.198	2.165	2.202	2.301	2.352	2.352	(41)	
	Ebn at Noon		747	793	819	801	827	832	821	794	778	757	721	708	(42)	
		Edh at Noon	81	105	124	156	145	138	140	139	124	98	79	72	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.21	2.05	3.29	4.57	5.55	6.16	6.20	5.51	3.85	2.63	1.59	1.01	(44)	
	RadStd		0.15	0.27	0.41	0.46	0.53	0.56	0.47	0.52	0.42	0.29	0.14	0.15	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.61	N/A	N/A	N/A	+0.64	+0.76	N/A	N/A	N/A	N/A	(46)
(47)	Regional (59 neighbors)	+0.67	N/A	N/A	N/A	+0.37	N/A	-111	-173	+118	+64	(47)

Nomenclature: See separate page